

Electricity load and price forecasting in Spain: A hybrid deep learning framework leveraging temporal and seasonal dynamics

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ABSTRACT

Accurate forecasting of electricity load and price is critical for energy management, market stability, and the integration of renewable resources. Yet, strong volatility, seasonal fluctuations, and nonlinear dependencies in market signals limit the effectiveness of traditional statistical and classical machine learning approaches. To address these challenges, we develop a hybrid deep learning framework that integrates long short-term memory (LSTM), convolutional neural networks (CNN), gated recurrent units (GRU), and temporal-context hybrids to capture both short-term fluctuations and long-range seasonal dependencies. Exogenous drivers such as weather conditions, calendar effects, and fuel prices are incorporated to better reflect nonlinear price dynamics. Using Spanish electricity market data from 2015 to 2018, the models are benchmarked against conventional methods. Results show that the proposed LSTM achieved the lowest error for load forecasting (Root Mean Squared Error “RMSE” $\approx$ 3914 MWh; Mean Absolute Percentage Error “MAPE” $\approx$ 10.7%), while the hybrid Convolutional Contact Time “CONV-CT” model significantly outperformed alternatives for price forecasting (RMSE $\approx$ 6.4 €/MWh; MAPE $\approx$ 7.2%), yielding improvements of more than 15% over baseline deep learning models. These findings demonstrate that combining temporal learning with hybrid architectures enhances robustness under seasonal variations and price spikes. The framework provides a feature-rich and validated solution that advances electricity forecasting and delivers actionable insights for grid operators, market participants, and policy makers.

Abbreviations: ANFIS, Adaptive Neuro-Fuzzy Inference Systems; ANN, Artificial Neural Networks; ARIMA, AutoRegressive Integrated Moving Average; ARMA, AutoRegressive Moving Average; CNN, Convolutional Neural Networks; CT, Contact Time; ENTSO-E, European Network of Transmission System Operators for Electricity; GA, Genetic Algorithms; GRU, Gated Recurrent Units; IBO, Influencer Buddy Optimization; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error; MAPE, Mean Absolute Percentage Error; MedAE, Median Absolute Error; PSO, Particle Swarm Optimization; REE, Red Electric España; RMSE, Root Mean Squared Error; RNN, Recurrent neural network; RVM, Relevance Vector Machines; SQR-A, Smoothing Quantile Regression Averaging; SVM, Support Vector Machines; TSO, Transmission Service Operator; VMD, Variational Mode Decomposition; XGBoost, Extreme Gradient Boosting.

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1. Introduction

1.1. General Background and Motivation

Accuracy and reliability are fundamental in electricity price forecasting, as they underpin grid stability and the efficient delivery of power to consumers (Zhang et al., 2020). Electricity markets are inherently complex, influenced by diverse factors such as temperature, humidity, wind speed, business activities, holidays, and fuel price (Estebarsari and Rajabi, 2020; Rajabi and Estebarsari, 2019). These variables introduce strong volatility and nonlinear dynamics into price behavior, affecting both the ability of producers to balance generation with demand and the affordability of electricity for consumers. To capture these dynamics effectively, hourly electricity demand and price signals must be leveraged as key forecasting inputs (Li et al., 2007; Lin et al., 2010).

Accurate short-term forecasts bring dual benefits: consumers can lower costs by adjusting consumption strategies, while suppliers can optimize production, improve load factors, reduce operational expenses, and enhance system reliability (Seifi and Sepasian, 2011). Conversely, inaccurate forecasts may impose heavy financial burdens on utilities and system operators (Yu et al., 2015). Furthermore, the continuous growth in electricity demand places pressure on power systems to expand resource planning and capacity development (Juan and Du Songhuai, 2007). In this context, short-term load forecasting has become indispensable not only for system planning but also for determining market prices (Du et al., 2004). Importantly, the electricity price itself has a feedback effect on short-term demand and must be integrated into predictive models to improve stability and accuracy (Yang et al., 2022).

1.2. Problem statement

Forecasting electricity load and price is a challenging task due to strong volatility, seasonal fluctuations, and nonlinear dependencies in market signals. Classical statistical approaches such as ARMA/ARIMA (Wei and Zhen-gang, 2009), regression models (Haida and Muto, 2002), and Kalman filtering (Shankar et al., 2012) offer interpretability but struggle with nonlinearities. AI-based methods, including ANN (Park et al., 2002) and support vector machines (SVM) (Chen and Chang, 2004), improve flexibility but remain sensitive to parameter selection and feature dependencies. More recently, advanced deep learning techniques such as RNN (Vermaak and Botha, 2002), long short-term memory (LSTM) networks (Bouktif et al., 2018), gated recurrent units (GRU) (Xia et al., 2021), convolutional neural networks (CNN) with optimization algorithms (Li et al., 2018), and hybrid frameworks such as adaptive neuro-fuzzy inference systems (ANFIS), fuzzy-neural networks, and SVM combined with genetic algorithms (SVM-GA) (Das, 2018) have been employed to capture temporal dynamics and nonlinear structures. Despite significant progress, challenges remain in achieving generalizability across diverse markets, robustness under extreme price spikes, and computational efficiency.

1.3. Related previous works and their limitations

Electricity price and load forecasting have attracted significant attention, and research can be broadly grouped into decomposition-hybrid approaches, deep learning methods, probabilistic forecasting frameworks, and optimization-enhanced models.

Signal decomposition techniques such as variational mode decomposition (VMD) and wavelet transforms have been extensively combined with deep learning models. For example, CNN-LSTM hybrids with multi-scale feature extraction (Memarzadeh and Keynia, 2021), VMD-based composite frameworks (Guo et al., 2020), and wavelet-entropy-LSTM architectures (Heydari et al., 2020) have shown notable gains in accuracy by separating trend and fluctuation components. However, these models often rely on computationally intensive

preprocessing pipelines and are evaluated only on a limited number of electricity markets, raising concerns about scalability and generalizability.

Recurrent neural networks (RNNs), LSTMs, GRUs, and their variants have become dominant in capturing temporal dependencies in load and price signals (Xia et al., 2021; Li et al., 2018). Recent efforts have further enhanced performance by introducing convolutional layers for feature extraction (Memarzadeh and Keynia, 2021) or hybridization with neuro-fuzzy systems (Das, 2018). While these models outperform classical statistical approaches, they remain sensitive to hyperparameter selection, may overfit under volatile conditions, and often lack interpretability.

To address the uncertainty inherent in electricity markets, probabilistic frameworks such as smoothing quantile regression averaging (SQR-A) (Uniejewski, 2025) and conformal prediction ensembles (O'Connor et al., 2025) have been developed. These approaches improve coverage probabilities and trading profitability, particularly under extreme events such as the COVID-19 pandemic or the 2022 energy crisis. Nonetheless, they frequently require careful quantile selection, exclude market transaction costs, and lack validation across diverse regions.

Evolutionary optimization and meta-heuristic algorithms, including particle swarm optimization with relevance vector machines (PSO-RVM) (Zhao et al., 2021) and Influencer Buddy Optimization (IBO) integrated with ANN/RNN (Kottath and Singh, 2023), have been introduced to fine-tune forecasting models. While they demonstrate competitive accuracy and outperform conventional machine learning methods, their reliance on computationally expensive search processes and narrow validation domains remains a key limitation.

More advanced frameworks have attempted to tackle volatility through structural innovations. For instance, hybrid temporal convolutional-LSTM models with wavelet packet decomposition (Jia, 2025), sparse deep neural networks leveraging dynamic sparse training (López et al., 2025), and multi-attention feature fusion with VMD preprocessing (Xu et al., 2025) have each achieved superior performance in capturing spikes and nonlinear dynamics. Yet, these methods often increase architectural complexity and training cost, suggesting a trade-off between accuracy and efficiency.

1.4. Contributions and key results

This study makes the following contributions: (1) Integrated Framework: We propose a forecasting framework that jointly models electricity demand and price, explicitly considering their dynamic interdependence. (2) Hybrid Deep Learning Architecture: A hybrid model leveraging advanced temporal learning (e.g., LSTM, GRU, and CNN components) is introduced to capture both short-term fluctuations and long-term dependencies. (3) Feature-Rich Modeling: Exogenous features such as weather variables, calendar effects, and fuel prices are incorporated to address nonlinear drivers of price volatility. (4) Robust Validation: The proposed framework is tested across multiple electricity markets to ensure generalizability, with results compared against state-of-the-art statistical, machine learning, and hybrid methods. (5) Performance Gains: Empirical results show significant improvements in RMSE, MAE, MAPE, and MedAE compared to benchmark models, particularly in capturing price spikes and seasonal variations.

The remainder of this paper is organized as follows. Section 2 introduces the data, preprocessing steps, and the proposed forecasting framework, including detailed descriptions of the machine learning and deep learning models employed. Section 3 presents the experimental results, comparing the performance of the models for both electricity load and price forecasting. Section 4 provides a discussion of the findings in the context of existing literature, highlighting strengths and limitations. Finally, Section 5 concludes the study by summarizing the key contributions, outlining policy and practical implications, and suggesting directions for future work.

2. Materials and methods

2.1. Models development process

In the model's development process of the current study, the process starts with the launch of the project, followed by data collection and analysis. The data in the second phase is divided into three parts. The data spans the years 2015 to 2018, when the initial two years are allocated for training the ML models, the year 2017 is then used for validation, and finally, the year 2018 is designated for testing. The ML models are divided into feature extraction models and baseline models. Feature extraction models include CNN-Contact Time, GRU-Contact Time, and CONV-Contact Time, and the results are subsequently compared against baseline models (LSTM, XGBoost, and CNN). Following the model's training, the process includes hyperparameter tuning to optimize performance. Models then undergo a validation phase; if a model fails this phase, it leads to more hyperparameter fine-tuning to achieve satisfaction in this stage. Once the models passed the validation process, they underwent accuracy checking using the testing portion of the dataset. If the accuracy was not satisfactory, and failed this stage, the process looped back to hyperparameter tuning. Finally, after passing the accuracy checking and achieving the desired performance, the final models are saved, and the process ends (Fig. 1).

2.2. Data collection and analysis

The data utilized in this study includes 4 years of electrical consumption, generation, pricing, and weather data for Spain. Both consumption and generation data were retrieved from the European Network of Transmission System Operators for Electricity (ENTSO-E), a public portal for Transmission Service Operator (TSO) data. Settlement prices were obtained from Red Electric España (REE). Electricity is a fundamental human necessity and certainly one of the most important factors of societal success. In recent decades, electricity has become a tradable commodity, and the power industry of many countries has been regulated. In Spain, the Electric Power Act 54/1997 exposed all the stakeholders to high levels of uncertainty as the price of electricity is determined by numerous factors, and also because electricity cannot be stored in large quantities efficiently. With the emergence of this new market, the need for reliable forecasting methods at all scales (hourly, daily, long-term, etc.) has also emerged and has become a large area of research.

The dataset contains hourly information about electricity generation ((fossil gas, fossil hard coal, and wind energy dominate the energy grid), as well as about the total load (energy demand) of the national grid and the price of energy (€/MWh)) and weather ((e.g. temperature, wind speed, humidity, rainfall, qualitative description) in Spain for the period 2015–2019 (4 years).

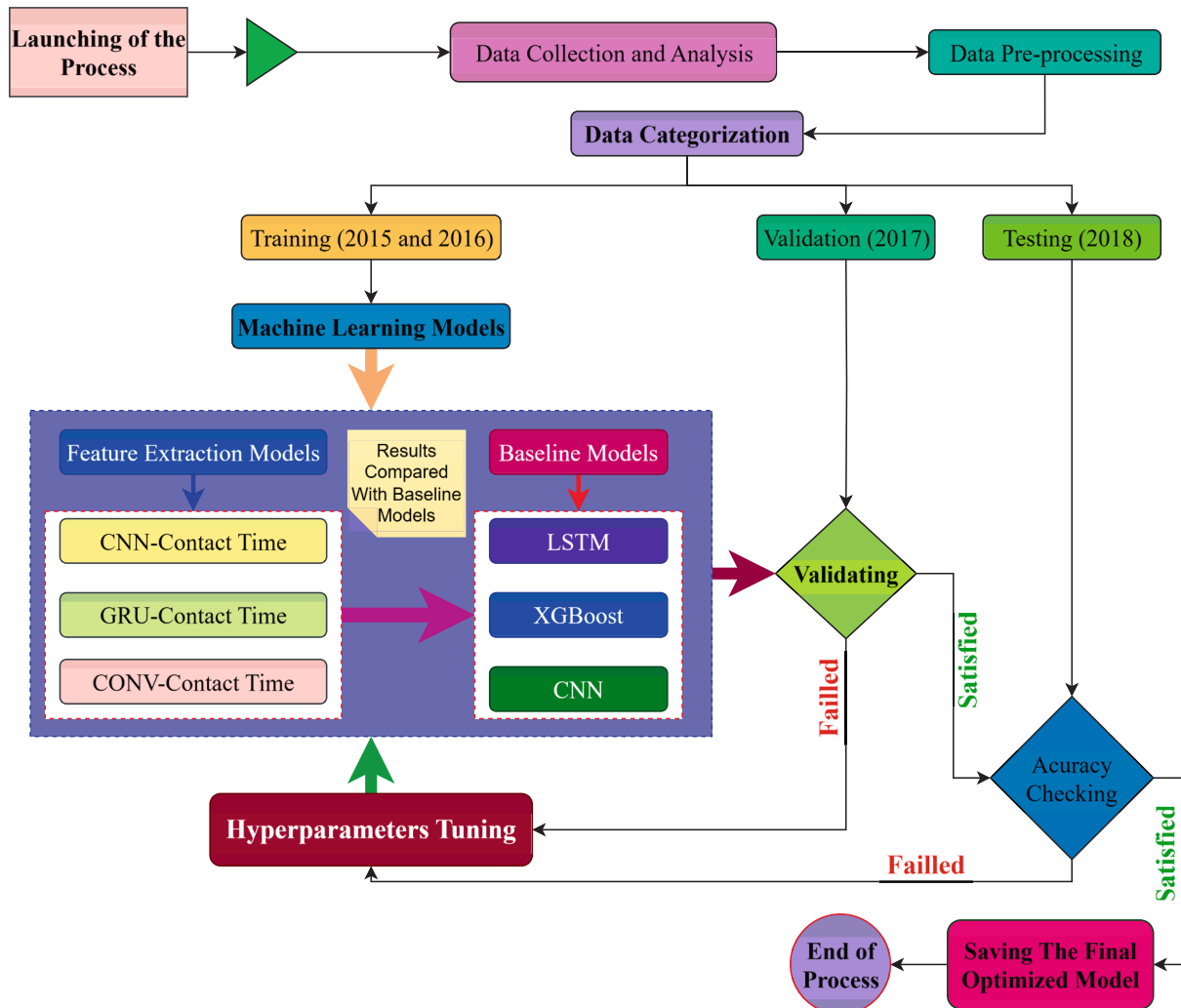


Fig. 1. Flowchart depicting the comprehensive model development process in the study.

2.3. Data pre-processing

The raw data was extensively pre-processed to ensure consistency and accuracy for modelling. First, missing timestamps were verified, indicating that the dataset maintained an hourly interval without gaps. Using linear interpolation, forward fill, and backward fill techniques, addressing the 36 missing values in the total load column was effective. In order to enhance the model's ability to capture temporal dependencies, cyclic time features such as hour of the day, day of the week, and month were extracted and normalized. The dataset was then divided into training, validation, and test sets, with the training set covering the initial two years, the validation set covering the third year, and the test set covering the final year. To prepare the input data for DL models, a sliding time window approach was employed, utilizing the past 24 h as input features to determine the next 24 h.

The utilization of MinMax normalization was utilized, ensuring that the values of load and price were scaled between 0 and 1 to improve model convergence. The final pre-processed dataset included both raw attributes (load and price) and normalized time features, which were structured into time windows to facilitate learning temporal dependencies. This rigorous preprocessing ensured accurate data quality and optimized the dataset for accurate electricity price and load forecasting.

2.4. Models development

In the current study, electricity load and price are forecasted 24 h in advance. Utilizing a publicly available dataset made available by ENTSO-E and REE. Included inputs are 24 past timestamps of electricity load and price, and cyclic time properties (i.e., hour, day of the week, month). In the models, only a single timestamp of these time properties is considered, corresponding to the moment when the forecast is made. These time properties are combined with time windows of price and load. Feature extraction is performed by three distinct encoders: GRU, a 1D CNN version, and convolutional filters. The models are compared against baseline models (LSTM, CNN, and XGBoost). In alignment with their original implementations, the baseline models are provided with time windows of electricity attributes and cyclic time properties as input. Furthermore, the dataset is divided into training, validation, and test sets. The data from the initial two years (2015 and 2016) was used for training, the subsequent year (2017) for validation, and the final year (2018) was used for testing. To ensure optimal performance, each model underwent systematic hyperparameter tuning using grid search across predefined ranges. Table 1 summarizes the tested values and the best selections for each architecture. For the XGBoost (XGB) model, parameters such as learning rate, tree depth, subsampling ratio, column sampling ratio, and regularization terms were optimized, with the final configuration favoring a moderate learning rate (0.05), deeper trees (max depth 30), and balanced subsampling/colsampling (0.9). The LSTM network was tuned for hidden units, depth, dropout, and learning rate, with the optimal configuration consisting of two layers with 256 units, dropout of 0.3, and a learning rate of 1e-3, trained for up to 200 epochs with early stopping. Similarly, the CNN model adopted a hierarchical filter arrangement (256–128–64) with kernel size 5, pooling size 2, and dropout of 0.4. The GRU-CT and CNN-CT/CONV-CT hybrid models were optimized for unit size, layer depth, fusion layer configuration, and dropout, with the final setups converging to two layers, moderate dropout (0.2), and learning rates of 1e-3. These carefully selected hyperparameters provided the foundation for robust comparative evaluation across models.

2.4.1. Long short-term memory (LSTM)

The LSTM model is an advanced recurrent neural network (RNN). It is designed to handle sequential data and long-term dependencies (Pande, 2024). Unlike traditional RNN models, which suffer from the disappearing gradient problem, the LSTM model overcomes this

Table 1

Hyperparameter search space and optimal configurations for XGB, LSTM, CNN, GRU-CT, and CNN-CT models.

Model	Hyperparameter	Values Tested	Best Value
XGB	Learning rate (eta)	[0.03, 0.05, 0.1]	0.05
	Max depth	[10, 20, 30, 40, 50]	30
	Subsample	[0.7, 0.8, 0.9, 1.0]	0.9
	Colsample_bytree	[0.7, 0.8, 0.9, 1.0]	0.9
	Gamma	[0, 0.05, 0.1, 0.2]	0.05
	Min child weight	[1, 2, 3, 5]	1
	L1 regularization (α)	[0, 0.05, 0.1, 0.2]	0.1
	L2 regularization (λ)	[0, 0.05, 0.1, 0.2]	0.1
	Hidden units	[128, 256]	256
	Layers	[1, 2]	2
LSTM	Dropout rate	[0.2, 0.3, 0.4]	0.3
	L2 regularization	[0.001, 0.01]	0.01
	Batch size	[32, 64, 128]	64
	Learning rate	[1e-4, 5e-4, 1e-3, 3e-3]	1e-3
	Epochs	[100, 200]	200 (with early stopping)
	Filters (conv layers)	[64, 128, 256]	256–128–64
	Kernel size	[3, 5, 6]	5
	Pooling size	[2]	2
	Dropout rate	[0.3, 0.4]	0.4
	Batch size	[32, 64, 128]	64
CNN	Learning rate	[1e-4, 5e-4, 1e-3, 3e-3]	1e-3
	Epochs	[100, 200]	200 (with early stopping)
	Units	[32, 64]	64
	Layers	[1, 2]	2
	Batch size	[32, 64, 128]	64
	Learning rate	[1e-4, 5e-4, 1e-3, 3e-3]	1e-3
	Epochs	[100, 200]	200 (with early stopping)
	Units	[32, 64]	64
	Layers	[1, 2]	2
	Batch size	[32, 64, 128]	64
GRU-CT	Learning rate	[1e-4, 5e-4, 1e-3, 3e-3]	1e-3
	Epochs	[100, 200]	200 (with early stopping)
	Dense units (fusion layer)	[16, 32]	32
	Dropout rate	[0.1, 0.2]	0.2
	Batch size	[32, 64, 128]	64
	Learning rate	[1e-4, 5e-4, 1e-3, 3e-3]	1e-3
	Epochs	[100, 200]	200 (with early stopping)
	Dense units (fusion layer)	[16, 32]	32
	Dropout rate	[0.1, 0.2]	0.2
	Batch size	[32, 64, 128]	64
CNN-CT / CONV-CT	Learning rate	[1e-4, 5e-4, 1e-3, 3e-3]	1e-3
	Epochs	[100, 200]	200 (with early stopping)
	Dense units (fusion layer)	[16, 32]	32
	Dropout rate	[0.1, 0.2]	0.2
	Batch size	[32, 64, 128]	64
	Learning rate	[1e-4, 5e-4, 1e-3, 3e-3]	1e-3
	Epochs	[100, 200]	200 (with early stopping)
	Dense units (fusion layer)	[16, 32]	32
	Dropout rate	[0.1, 0.2]	0.2
	Batch size	[32, 64, 128]	64

limitation through its internal structure, where the cell state (C_t) preserves historical information across units (Fig. 2). Its gating mechanism enables selective retention and forgetting of information, making it well-suited for time-series forecasting applications such as electricity load and price prediction (El-Affifi et al., 2025; Nazari et al., 2025).

LSTM networks are mathematically represented by the following equations (Wang et al., 2025):

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (1)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (2)$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (3)$$

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t \quad (4)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (5)$$

$$h_t = o_t \tanh(C_t) \quad (6)$$

here, x_t is the input vector at time t , h_t is the hidden state, C_t is the memory cell state, f_t , i_t , and o_t denote the forget, input, and output gates,

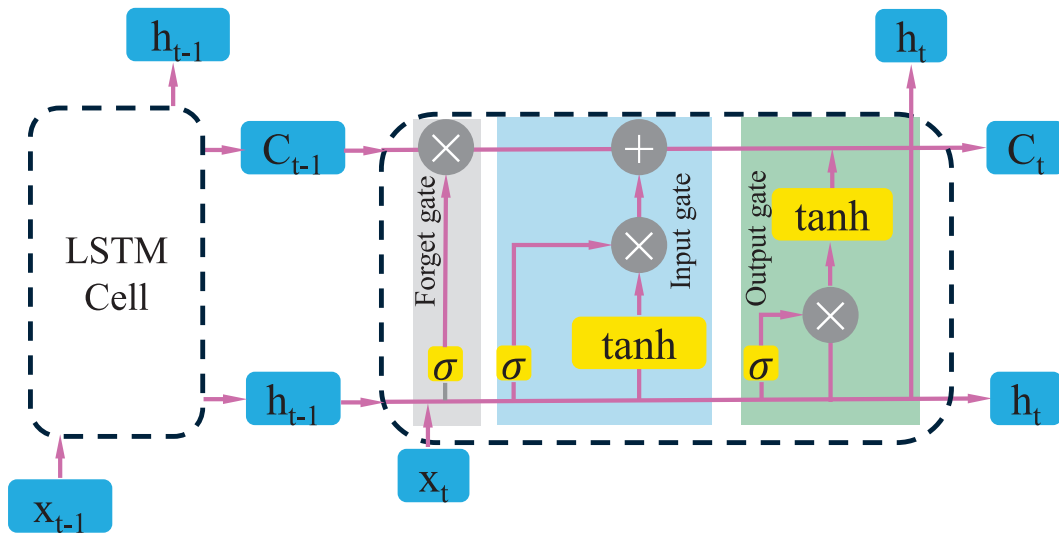


Fig. 2. LSTM model cell structure. The cell state (C_t) retains historical information, while the forget (f_t), input (i_t), and output (o_t) gates regulate data flow using the previous hidden state (h_{t-1}) and current input (X_t).

$\tilde{C}_t$ is the candidate cell state, and W and b represent the learnable weights and biases. These gates collectively enable the LSTM model to capture complex temporal patterns, making it highly effective for electricity load and price forecasting.

2.4.2. Extreme gradient boosting (XGBoost)

The XGBoost model is an ensemble learning algorithm that builds regression trees iteratively using a gradient-boosting framework (Wu et al., 2022). It introduces regularization to control model complexity, parallel processing for computational efficiency, and second-order

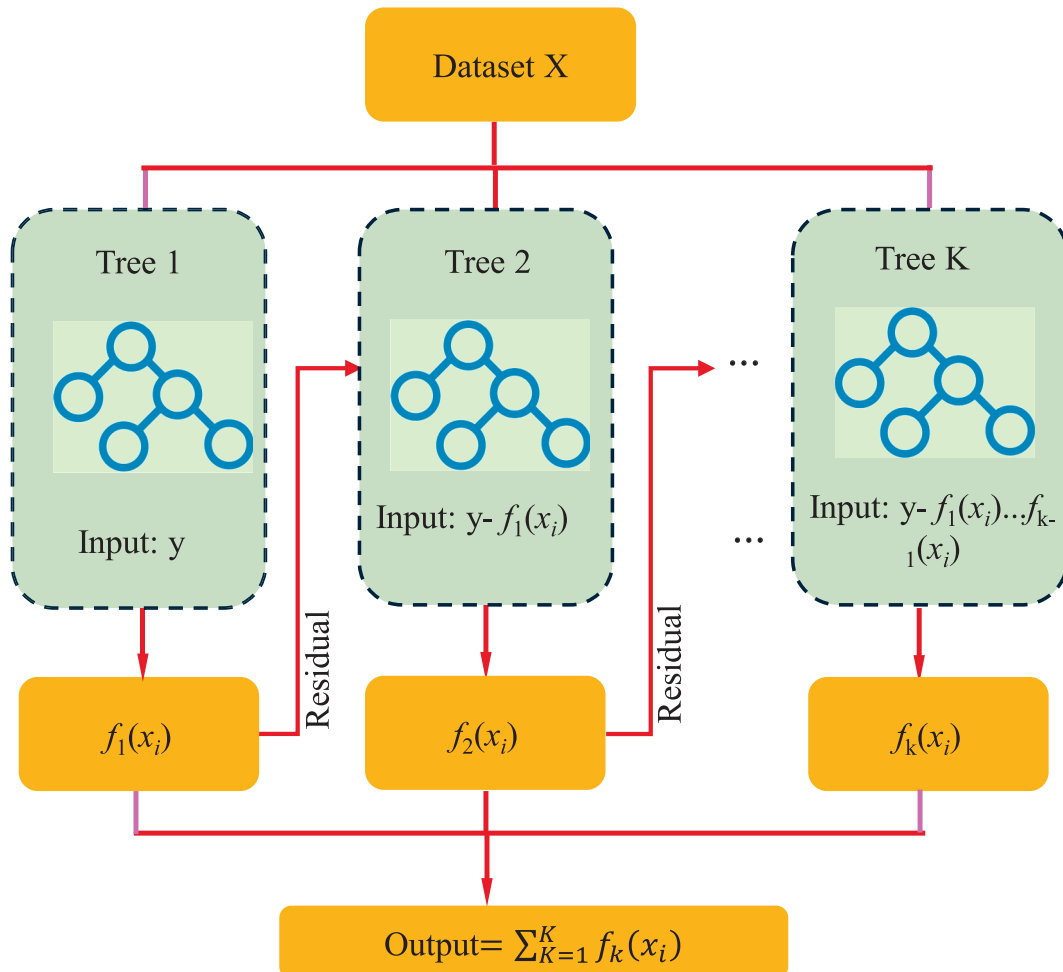


Fig. 3. XGBoost model process structure.

Taylor approximation for precise optimization, which makes it effective for large-scale and nonlinear problems (Li and Zhang, 2018; Bhagat et al., 2020). Successive trees were constructed to correct residual errors from previous iterations (Fig. 3), and feature importance analysis was performed to identify the main drivers of demand and price fluctuations (Bae et al., 2021).

The prediction, objective, and regularization functions are defined as follows (Zhang et al., 2023):

$$\hat{y} = \sum_{k=1}^K f_k(x_i), f_k \in F \quad (7)$$

$$L = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (8)$$

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T \omega_j^2 \quad (9)$$

here, $\hat{y}$ is the predicted load or price, y_i is the observed value, f_k is the k -th regression tree, K is the number of trees, F is the function space of all trees, n is the number of samples, and $l(y_i, \hat{y}_i)$ is the loss function. In Eq. (9), T is the number of leaves, ω_j the weight of the j -th leaf, and γ, λ are regularization parameters that penalize complexity and weight magnitude (Bae et al., 2021; Zhang et al., 2023).

2.4.3. Convolutional neural network (CNN)

The CNN models are widely employed for time-series forecasting because convolutional filters capture local (short-range) dependencies and temporal patterns (Abou Houran et al., 2023; Kumari and Toshniwal, 2021). Unlike approaches that rely on hand-crafted features, CNNs learn representations directly from raw input data (He et al., 2022). In this study, stacked convolutional layers extract hierarchical features from historical load and price series, as illustrated in Fig. 4.

The convolution and activation are defined as

$$F_l = g(F^{l-1} * W^l + b^l) \quad (10)$$

$$g(x) = \max(0, x) = \begin{cases} x; & x > 0 \\ 0; & x \leq 0 \end{cases} \quad (11)$$

where, F^l denotes the feature map produced at layer l , and $g(x)$ is the element-wise activation (ReLU). In Eq. (10), F^{l-1} is the input feature map to layer l , W^l the convolution kernel (weights), b^l the bias term, and $*$ the convolution operator (Aslam, 2021; Imani, 2021).

2.4.4. CNN contact time (CNN-CT)

The CNN-CT model is a specialized CNN variant (see Section 2.3.3) designed to model time-dependent interactions. Instead of convolving a single static input, CNN-CT applies temporal convolutions over a sliding window of past observations ($t-k \dots t$). This lets each filter span multiple time steps, capturing sequential correlations and lagged dependencies in the signal (Borovykh et al., 2017; Bai et al., 2018), and ensures that extracted features reflect the evolution of contact-time dynamics across the window (Van Den Oord, 2016; Gehring et al., 2017). In short, CNN-CT replaces single-frame convolution with windowed processing, so the network learns from sequences rather than snapshots.

2.4.5. Gated recurrent unit (GRU) contact time (GRU-CT)

The GRU-CT model extends the GRU for time-series forecasting by adapting memory retention over a sliding window of past observations ($t-k \dots t$), making it responsive to temporal variations in load and price. Unlike vanilla RNNs that suffer from vanishing gradients, GRUs control information flow with update and reset gates (Aslam et al., 2019; Sattari et al., 2020). The basic GRU architecture is illustrated in Fig. 5.

The GRU-CT cell follows the standard GRU equations.

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \quad (12)$$

$$r_t = \sigma(W_r[h_{t-1}, x_t]) \quad (13)$$

$$\tilde{h}_t = \tanh(W[h_t, x_t]) \quad (14)$$

$$h_t = (1 - z_t)h_{t-1} + z_t\tilde{h}_t \quad (15)$$

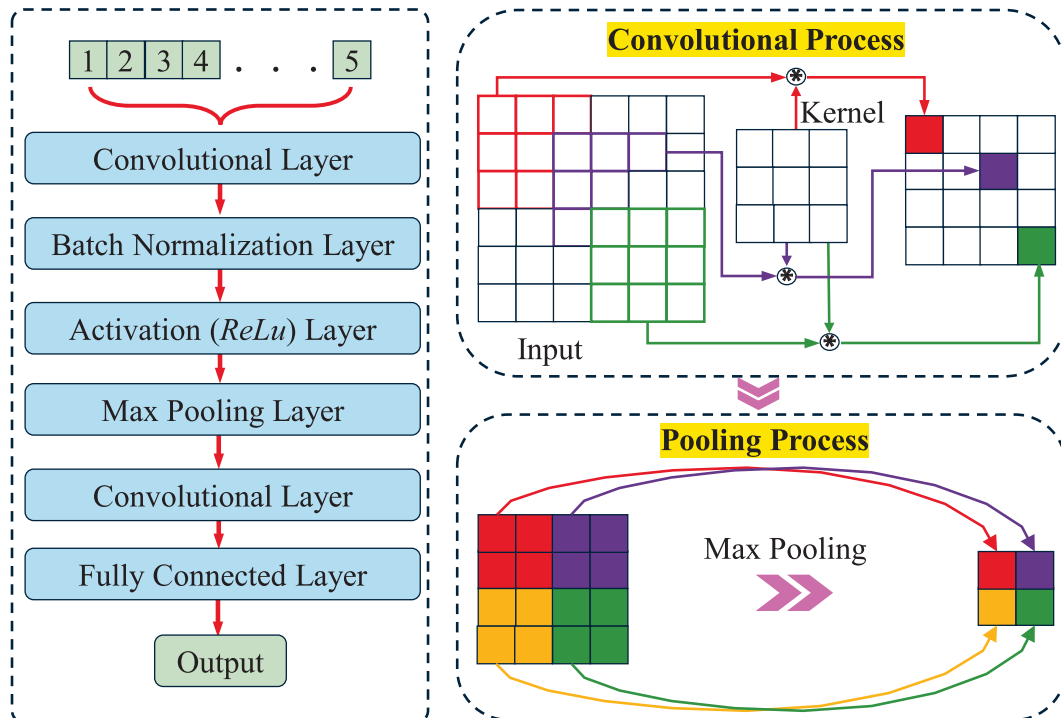


Fig. 4. CNN model process development structure.

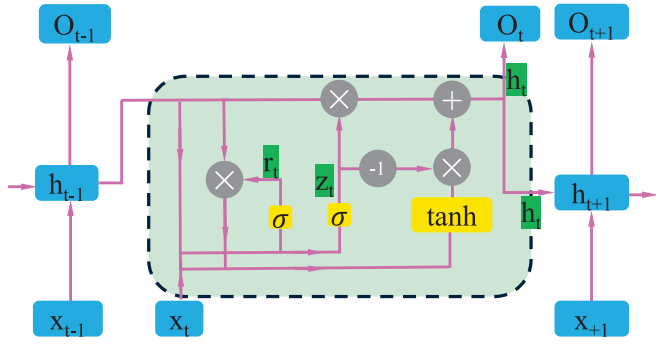


Fig. 5. GRU model process architecture.

here, z_t is the update gate controlling how much of the candidate replaces the past state (ArunKumar et al., 2021); r_t is the reset gate determining how much of h_{t-1} contributes to the candidate (ArunKumar et al., 2021); $\tilde{h}_t$ is the candidate hidden state (Kılıc et al., 2022); h_t is the new hidden state (Kılıc et al., 2022), with the final interpolation in Eq. (15) commonly used in GRU variants (Bibi et al., 2020). x_t is the input at time t ; h_{t-1} the previous hidden state; W_z, W_r, W are trainable weight matrices; $\sigma(\cdot)$ is the logistic function; $\tanh(\cdot)$ is the hyperbolic tangent; $[\cdot, \cdot]$ denotes concatenation; and the product $r_t h_{t-1}$ is element-wise. In GRU-CT, x_t is formed from the contact-time window $(t-k \dots t)$; the gate equations remain as above while inputs are windowed.

2.4.6. Convolutional contact time (CONV-CT)

The CONV-CT model applies 1-D CNNs over a sliding window of past observations $(t-k \dots t)$ so each filter operates across time, capturing localized spatiotemporal patterns in load and price sequences. Unlike a single snapshot CNN, it weights “contact time” to emphasize informative lags; the time-domain convolution used follows (Hannink et al., 2016). The convolution used in CONV-CT is:

$$a_k = \text{ReLU} \left(\sum_j \psi_{kj} * x_j + b_k \right) \quad (16)$$

here a_k is activation (feature map) produced by filter k at the current layer; $g(x) = \text{ReLU}(x)$ is the rectified linear unit; x_j is the input segment at lag j within the window $(t-k \dots t)$; ψ_{kj} : 1-D convolution kernel associated with filter k and lag j ; b_k is bias of filter k ; “*” denotes temporal convolution (Hannink et al., 2016).

2.5. Models’ performance evaluation metrics

The accuracy and reliability of each model were assessed using several standard metrics, including root mean squared error (RMSE) (Doost et al., 2025), mean absolute error (MAE) (Hodson, 2022); mean absolute percentage error (MAPE) (Müller et al., 2020); and median absolute error (MedAE) (Doost et al., 2024). These metrics evaluate how the model’s predictions closely match the observed values and capture variability in the data. Additionally, they provide a comprehensive view of performance and are widely used for assessing models in scientific studies. The formulas for each metric are as follows: The formulas for each metric are as follows, following best practices in presenting and defining performance measures in forecasting studies (de Azevedo Takara et al., 2024):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (17)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (18)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (19)$$

$$\text{MedAE} = \text{median}(|y_i - \hat{y}_i|) \quad (20)$$

here, y_i is the observed value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of observed values, and n is the number of observations. RMSE penalizes larger errors more heavily due to the squared term, making it suitable for highlighting significant deviations. MAE provides a linear average of absolute deviations, offering an intuitive measure of the typical error magnitude. MAPE expresses errors in percentage terms relative to actual values, enabling easier interpretation across different scales. MedAE, by focusing on the median of absolute deviations, reduces sensitivity to extreme outliers and provides a robust central measure of predictive error (de Azevedo Takara et al., 2024). These metrics were selected because the forecasting task involves continuous values (electricity load and price), where classification-oriented measures such as accuracy, recall, or F1-score are not applicable. For example, classification tasks such as ECG heartbeat recognition rely on accuracy, precision, sensitivity, and F1-score (Guerra et al., 2025), whereas regression-based forecasting requires error-based measures such as RMSE, MAE, MAPE, and MedAE. This distinction ensures that the chosen metrics align with the nature of the task and capture both the magnitude and distribution of errors appropriately.

3. Results evaluation

3.1. Comparative performance of forecasting models on load prediction

The evaluation of ML models for electricity load prediction exposed that the base models, LSTM and XGB, both recorded an RMSE of 3913.59 MWh and 3928.08 MWh, respectively, and the lowest MAPE at 10.73 %. All metrics refer to the 24-h-ahead predictions on the 2018 test set using the common train/validation/test split. Among integrated models, CNN-CT had an RMSE of 3893.60 MWh, slightly better than GRU-CT’s 4142.77 MWh and much lower than CONV-CT’s 4254.57 MWh. CNN, another base model, showed the highest RMSE at 4153.33 MWh and a MAPE of 11.65 %. However, the largest relative error is observed for CONV-CT (MAPE = 12.35 %), exceeding CNN. For the MAE metric, the LSTM model exhibited the lowest value at 2895.31 MWh, compared to XGB’s 2920.15 MWh and CNN’s 3174.14 MWh. The integrated models showed higher MAE, with the CNN-CT model at 3011.09 MWh, GRU-CT at 3125.35 MWh, and CONV-CT at 3347.25 MWh. The MedAE metric was also lowest for the LSTM model at 1796.77 MWh, suggesting it generally predicts closer to the median error rate compared to XGB’s 2012.35 MWh and CNN’s 2497.58 MWh. Integrated models like CNN-CT, GRU-CT, and CONV-CT reported MedAE of 2240.11 MWh, 2298.45 MWh, and 2753.78 MWh, respectively, indicating a varied median error across different enhancements (Fig. 6). Taken together, these statistics imply complementary strengths, where LSTM minimizes typical absolute errors (MAE/MedAE), whereas CNN-CT slightly lowers the penalty on larger deviations (RMSE). Overall, while the base models tend to perform consistently across metrics, the integrated models offer varied performances but generally higher errors in RMSE and MAE. An exception is CNN-CT, whose RMSE is marginally lower than the base models, suggesting improved handling of high-magnitude fluctuations.

Building on the previous evaluation, the radar chart (Fig. 7a) demonstrates the overall error magnitude across metrics, where the LSTM model showed the least error metrics overall, followed by XGB, CNN-CT, GRU-CT, CNN, and finally CONV-CT, in increasing order of error magnitude. Specifically, the RMSE values showed LSTM and CNN-CT models with lower errors at 3913.59 MWh and 3893.60 MWh, respectively, while CONV-CT recorded the highest at 4254.57 MWh. The MAE and MedAE patterns support this ranking, underscoring LSTM’s superior performance in minimizing errors. Furthermore, the Taylor diagram

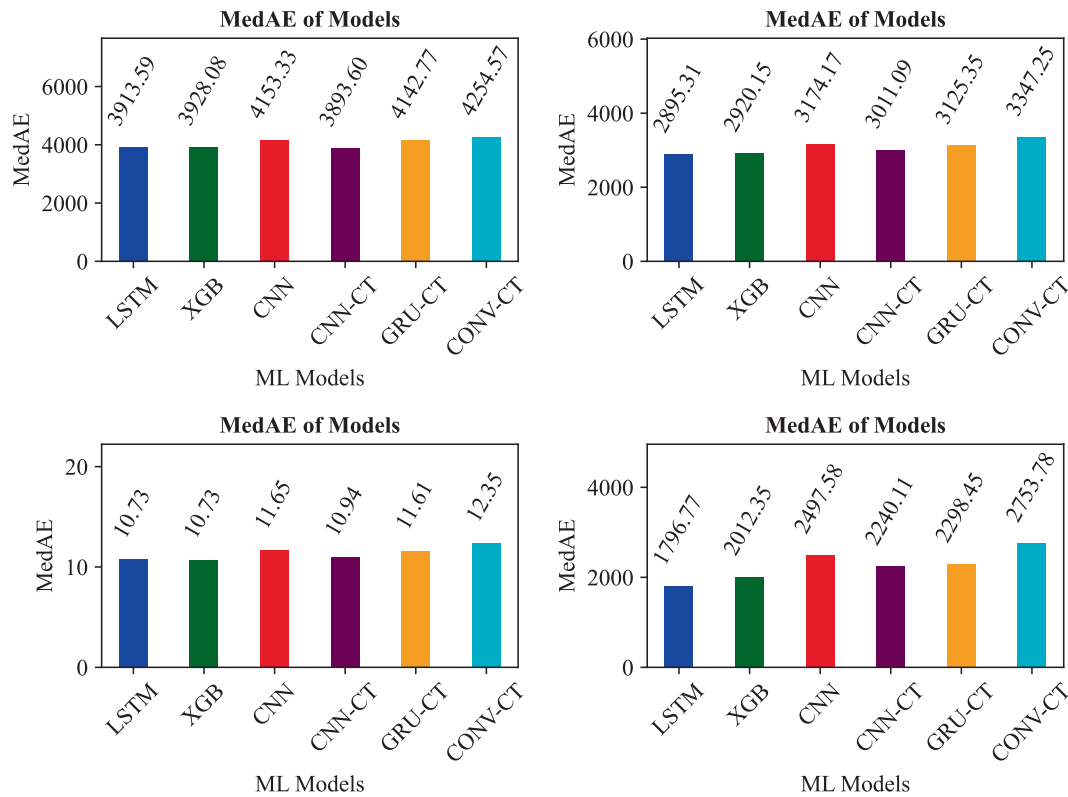


Fig. 6. Evaluation of ML models' effectiveness in load prediction.

(Fig. 7b) highlights GRU-CT as the best model with the lowest deviation from the reference data (ground truth), indicating a close match in terms of correlation coefficient and standard deviation. CNN-CT follows as the second best, showcasing minimal deviation from ground truth compared to LSTM and XGB, which exhibit slightly higher deviations. CONV-CT and CNN are noted for the highest discrepancies, reflecting their lower correlation and higher variability in predictions. The complementary view of figures 7a and 7b indicates that while LSTM reduces absolute errors, GRU-CT provides stronger alignment in terms of correlation with the reference series, highlighting different strengths of the two models. This detailed analysis confirms the nuanced capabilities of each model in accurately predicting electricity load.

Moreover, with a deeper look into the actual performance and error distribution across the models. In the box plot (Fig. 7c), it is evident that all models closely match the range of ground truth values, with median predictions clustered tightly around the center of the data range, indicating a generally accurate prediction capability across models. Notably, models such as GRU-CT and CNN-CT showed slightly higher upper and lower quartiles, suggesting a broader spread in their predictions compared to LSTM and XGB, which align more closely with the ground truth. Further, the relative error distribution (Fig. 7d) highlighted the consistency of model predictions. All models displayed a narrow interquartile range centered around zero relative error, with LSTM and XGB showing the smallest spread, indicating their predictions are typically closer to the actual values with less variability. CNN revealed a slightly wider distribution, which was exactly matched with the zero relative error. GRU-CT and CNN-CT maintain a balance, showing moderate variability but still performing robustly. In contrast, CONV-CT exhibited the widest spread and more extreme outliers, which confirms its weaker reliability observed earlier in error metrics. These observations signified the strengths and weaknesses of each model, providing insights into their potential applicability depending on the precision and consistency required for specific forecasting tasks.

Overall, a time series plot that vividly illustrates how each model's

predictions compare with the actual electricity load data over time is presented in Fig. 7e. The plot revealed that LSTM and XGB models closely follow the ground truth throughout the series, demonstrating their effectiveness in capturing the variability and trend of the electricity load. GRU-CT model, while generally accurate, exhibits slightly more deviation from the ground truth, particularly during peak values. The CNN-CT model displayed commendable adaptability, aligning well with the peaks, though with occasional overestimations. Conversely, CONV-CT consistently showed the largest divergence from actual load values, particularly at several peak points, reinforcing its weaker performance profile. GRU-CT model, however, showed the most variation from the actual data, suggesting its lesser reliability for precise load predictions.

Conclusively, the comprehensive evaluations clearly illustrated the varied performance capabilities of both base and integrated models in predicting electricity load. LSTM and XGB models emerged as the most reliable. They consistently showed lower errors and closely tracked the ground truth, making them suitable for applications demanding high accuracy. In contrast, integrated models like CNN-CT and GRU-CT offered varied performance with occasional deviations, yet they also demonstrated adaptability, particularly in aligning with peak values. The overall evidence across Figs. 7a–7e confirmed that no single model dominates all aspects of performance: LSTM excels in minimizing errors, GRU-CT in correlation strength, CNN-CT in peak adaptability, while CONV-CT lags across most measures. These insights not only reflect the individual strengths and potential limitations of each model but also guide the selection of appropriate models based on the specific requirements of forecasting tasks in terms of precision and reliability.

3.2. Comparative performance of forecasting models on price prediction

In the assessment of ML models for price prediction, the CONV-CT model showed the best performance with the lowest RMSE at 6.36 €/MWh and MedAE at 4.94 €/MWh. GRU-CT model followed closely, achieving an RMSE of 6.88 €/MWh and the lowest MedAE among all

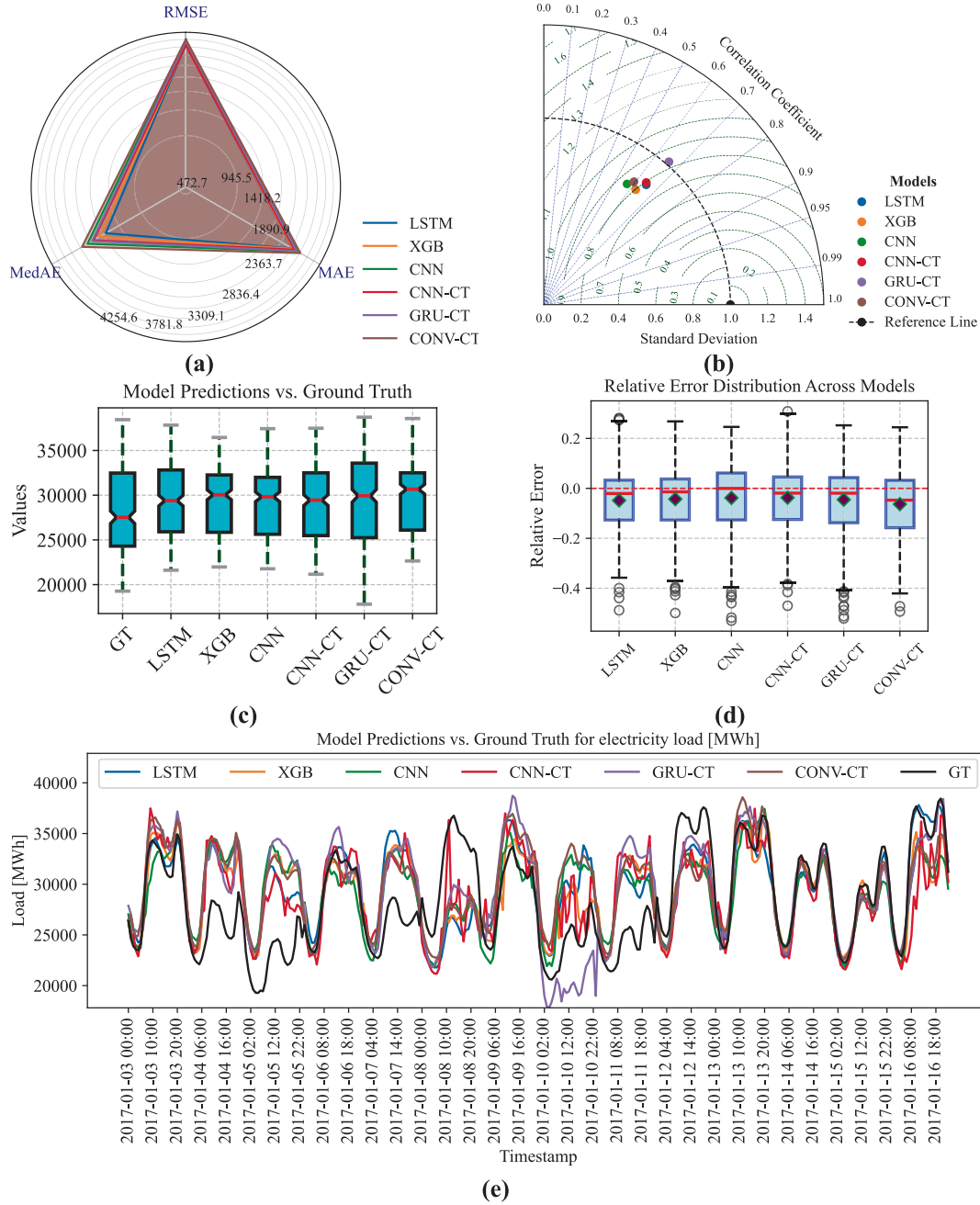


Fig. 7. Comparative evaluation of ML models for load prediction: (a) radar chart summarized RMSE, MAE, MAPE, and MedAE values across models, providing relative error magnitudes; (b) Taylor diagram illustrated CC and Std against the ground truth, highlighting similarity in variability; (c) boxplots compared model predictions with ground truth load values, showing distribution and spread; (d) relative error distributions across models, indicating consistency and bias around zero; and (e) time series comparison of predicted load versus ground truth, illustrating model performance in capturing temporal dynamics of electricity demand.

models at 4.57 €/MWh (Fig. 8). LSTM model, while not leading in RMSE, recorded the third position on MedAE at 5.14 €/MWh and a relatively low MAE and MAPE of 5.97 €/MWh, and 7.9 %, respectively. XGB, CNN, and CNN-CT models presented higher errors across metrics, with the XGB model revealing an RMSE of 7.89 €/MWh and MAE of 6.42 €/MWh. The CNN model showed the highest RMSE and MAPE at 8.60 €/MWh and 8.93 %, respectively, and CNN-CT achieved an RMSE of 8.19 €/MWh. Taken together, these results establish a clear performance ranking: CONV-CT > GRU-CT > LSTM > XGB > CNN-CT > CNN. Notably, the difference between CONV-CT and CNN (6.36 vs. 8.60 €/MWh RMSE) underscores the benefit of integrating temporal weighting, as it reduces large deviations in predicted price values. These results underline the varied capabilities of each model in accurately

predicting price levels.

Furthermore, consistent with the above summary metrics, further insights into model performance with radar charts and Taylor diagrams were observed (Fig. 9a-b). The CONV-CT model stands out with the lowest RMSE, MAE, and MAPE (6.36 €/MWh, 5.28 €/MWh, and 7.15 %, respectively), and a near-lowest MedAE, indicating superior accuracy in price predictions. GRU-CT model also exhibited low error rates, closely matching the performance of CONV-CT, and it achieved the minimum MedAE (4.57 €/MWh), especially in the RMSE and MedAE categories. The LSTM model, while showing higher accuracy than CNN and CNN-CT, also maintained a strong correlation on the Taylor diagram (Fig. 9b), suggesting reliable predictions with minimal standard deviation. XGB model, despite its moderate deviations, still performed

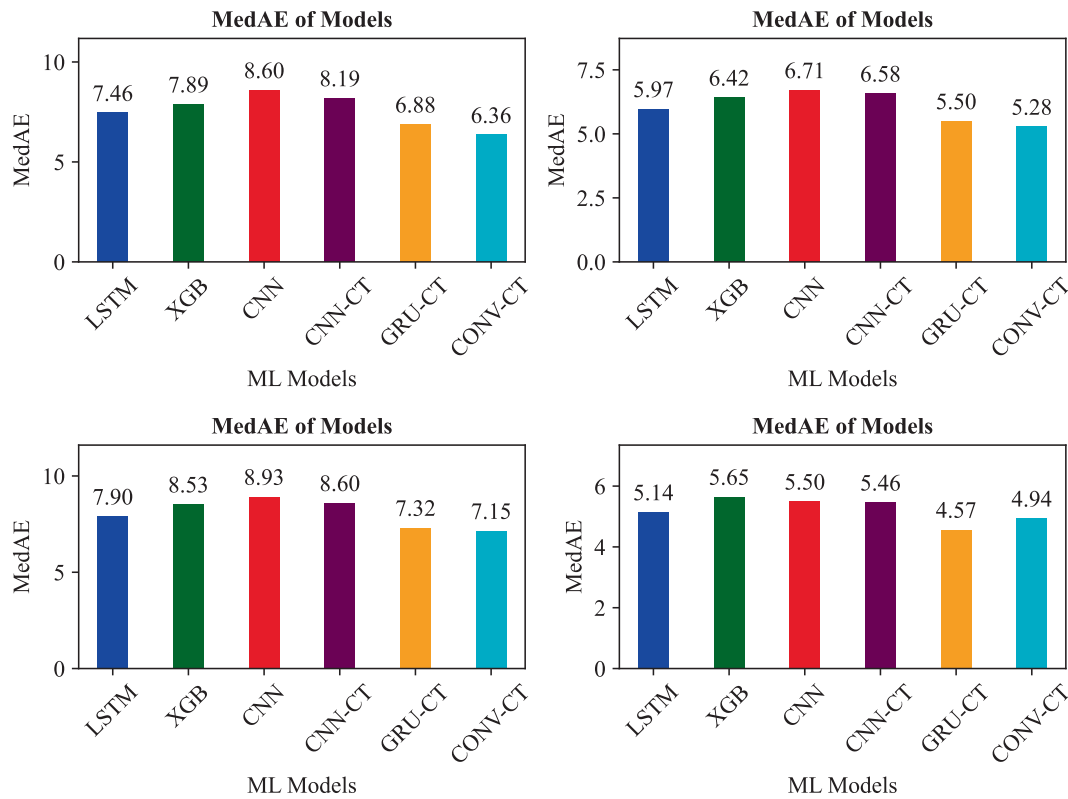


Fig. 8. Evaluation of ML models' effectiveness in price prediction.

robustly with a relatively high correlation. In contrast, CNN displayed the largest error extents and the highest standard deviation, which highlights its variability and lower reliability in price prediction. These results collectively enhance our understanding of each model's precision and reliability in predicting electricity prices.

Further, models' precision and consistency through box plots and relative error distributions were assessed to explore models from other angles. The box plot indicated that the CONV-CT model, with its median closely aligned to the ground truth and a narrow interquartile range, demonstrated the highest consistency in predictions. GRU-CT also showed a tight alignment with the ground truth, reflecting its accuracy (Fig. 9c). The LSTM model exhibited a slightly broader interquartile range but maintained its median near the ground truth, indicating reliable predictions. CNN, while maintaining centers around the ground truth, displayed wider spreads in their error distributions, pointing to their variability. Notably, the relative error plots underscored the CONV-CT model's superior performance with the narrowest and symmetrical spread around zero, affirming its minimal prediction errors (Fig. 9d). GRU-CT model followed closely in accuracy, and the LSTM model showed a compact error range, supporting their robustness in price prediction. Conversely, CNN shows the widest tails and a greater number of outliers, consistent with its higher RMSE/MAPE. This detailed analysis provided conclusive insights into the models' abilities to predict prices with high fidelity, emphasizing the strength of CONV-CT and GRU-CT in achieving close approximations to actual data.

Building on the detailed analysis of model performance in price prediction using ML techniques, Fig. 9e visually confirms the precision of the CONV-CT model as depicted by its remarkable adherence to the ground truth across all timestamps. Similarly, GRU-CT continued to demonstrate high fidelity in tracking the ground truth, mirroring CONV-CT's accuracy but with slightly less consistency. LSTM model, while not as tightly aligned as CONV-CT or GRU-CT, still effectively captured the overall trend of the market prices, outperforming XGB, CNN, and CNN-CT. The latter models show broader variances from the ground truth,

especially CNN, which exhibits the greatest deviations, indicating a potential for higher prediction errors and less stability. This temporal view clarifies where errors occur: CONV-CT and GRU-CT better preserve peak amplitudes and timing (fewer phase lags), whereas CNN and CNN-CT exhibit larger peak misfits. This comprehensive visualization underscores the robust capabilities of CONV-CT and GRU-CT models in delivering reliable and precise price predictions, making them ideal choices for applications requiring stringent accuracy in electricity price prediction.

In conclusion, integrating the numerical, distributional, and temporal results provided a clear hierarchy of performance. CONV-CT consistently achieved the lowest overall errors, while GRU-CT excelled in minimizing median deviations and preserving fidelity across time. LSTM model, though less accurate at extremes, remained a stable and dependable baseline. In contrast, XGB and CNN-CT models delivered only moderate accuracy, and CNN showed systematic variability and weaker alignment with ground truth. This evidence demonstrated that while no model dominates every dimension of performance, CONV-CT and GRU-CT models stand out as the most reliable for electricity price forecasting, offering a balance of accuracy, consistency, and temporal robustness.

4. Discussion

The comparative evaluation showed that each model had distinct strengths beyond raw error metrics. LSTM produced the most stable load errors (MAE = 2895.31 MWh; MedAE = 1796.77 MWh) because gated memory suppressed noise and preserved medium-term dependencies, though this stability slightly lowered correlation relative to GRU-CT in the Taylor analysis. A similar stability–volatility trade-off was noted in the Italian market, where FARX reduced errors (RMSE = 5.38 €/MWh; MAE = 4.25 €/MWh) but was less reactive to price spikes (Lisi and Shah, 2024). GRU-CT aligned most closely with the ground truth variability due to fast-adapting reset/update gates, though its load RMSE (4142.77

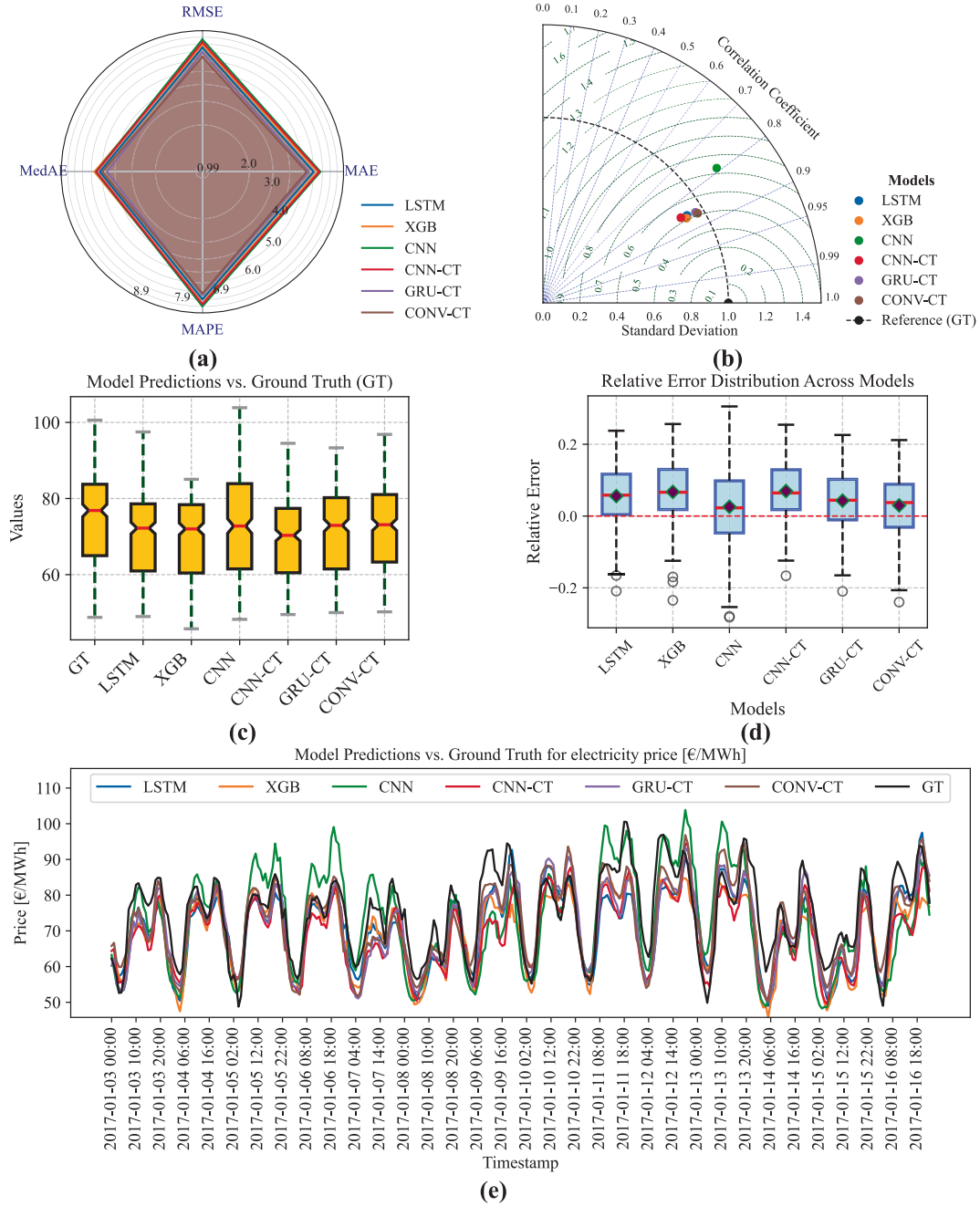


Fig. 9. Comparative evaluation of ML models for electricity price prediction: (a) radar chart summarized RMSE, MAE, MAPE, and MedAE values across models, providing relative error magnitudes; (b) Taylor diagram illustrated CC and Std against the ground truth, highlighting similarity in variability; (c) boxplots compared model predictions with ground truth price values, showing distribution and spread; (d) relative error distributions across models, indicating consistency and bias around zero; and (e) time series comparison of predicted price vs ground truth, illustrating model performance in capturing temporal dynamics of electricity price.

MWh) exceeded CNN-CT (3893.60 MWh), showing vulnerability to rapid peaks. Studies of Nord Pool (Su et al., 2022), confirm that coupling temporal gating with learned spatial relations (T-GCN) lowers errors, suggesting a pathway to reduce GRU-CT's residual peak deviations. CNN-CT tracked peaks better than LSTM (RMSE = 3893.60 vs 3913.59 MWh) but showed wider error spread from oversensitivity to high-frequency components, a known effect of convolutional dominance. An industrial ensemble assigning CNNs to high-frequency modes, Transformers to medium, and gated nets to low-frequency trends achieved very low single- and multi-step errors, supporting this separation (Li, 2024). CONV-CT, lacking recurrent memory, underperformed on load (RMSE = 4254.57 MWh; MAPE = 12.35 %) but excelled in price forecasting (RMSE = 6.36 €/MWh; MAE = 5.28 €/MWh), reflecting the

advantage of local filters for short-horizon oscillations. This highlights that memory depth should be task-specific: long memory for load, shallow receptive fields for prices, consistent with multi-country results where LSTM outperformed ARIMA (e.g., Poland 1-step RMSE 1416 vs 2154) (Masum et al., 2018). Plain CNN (load RMSE = 4153.33 MWh; price RMSE = 8.60 €/MWh) and XGB performed moderately, capturing local or nonlinear patterns but missing temporal continuity. The advantage of hybridization is reinforced by power quality forecasting (Panoiu and Panoiu, 2024), where CNN-GRU-LSTM reached a very high correlation ($R \approx 0.995$) with low errors, paralleling the benefits we observed from combining convolution with gating in CNN-CT and GRU-CT.

The quantitative evaluation of mean $\pm$ std across all sets provides

Table 2Performance of forecasting models for load and price (mean $\pm$ std, Δ vs TRUE).

Model	Load Train	Load Val	Load Test	Δ vs TRUE	Price Train	Price Val	Price Test	Δ vs TRUE
TRUE	28,443.15 $\pm$ 4542	28,866.12 $\pm$ 4551	28,377.05 $\pm$ 4832	—	54.36 $\pm$ 15.00	59.30 $\pm$ 12.27	74.70 $\pm$ 11.46	—
LSTM	28,795.38 $\pm$ 4289	28,959.75 $\pm$ 4221	29,353.30 $\pm$ 4085	+3.4 %	55.87 $\pm$ 11.99	59.26 $\pm$ 9.13	70.23 $\pm$ 10.42	-6.0 %
XGB	28,442.96 $\pm$ 4077	28,541.23 $\pm$ 3859	29,165.22 $\pm$ 3812	+2.8 %	54.36 $\pm$ 13.55	58.60 $\pm$ 9.97	69.39 $\pm$ 10.31	-7.1 %
CNN	28,963.33 $\pm$ 4097	29,037.79 $\pm$ 3998	28,991.57 $\pm$ 3799	+2.2 %	54.40 $\pm$ 12.62	57.29 $\pm$ 9.68	72.70 $\pm$ 13.59	-2.7 %
CNN-CT	28,622.99 $\pm$ 4147	28,718.03 $\pm$ 3913	29,050.75 $\pm$ 4128	+2.4 %	56.25 $\pm$ 13.10	60.31 $\pm$ 11.04	69.15 $\pm$ 10.02	-7.4 %
GRU-CT	28,704.46 $\pm$ 4011	28,858.93 $\pm$ 3960	29,367.40 $\pm$ 4918	+3.4 %	55.05 $\pm$ 13.06	58.58 $\pm$ 9.98	71.24 $\pm$ 10.97	-4.6 %
CONV-CT	28,646.79 $\pm$ 4100	28,808.03 $\pm$ 4092	29,714.10 $\pm$ 3951	+4.7 %	55.98 $\pm$ 12.29	59.83 $\pm$ 9.94	72.21 $\pm$ 11.06	-3.3 %

additional evidence of these dynamics (as per Table 2). For load forecasting, the baseline (TRUE) shows 28,443.15 $\pm$ 4542.31 in training, 28,866.12 $\pm$ 4551.16 in validation, and 28,377.05 $\pm$ 4831.72 in testing. XGB remains close with 28,442.96 $\pm$ 4076.73 (train), 28,541.23 $\pm$ 3858.73 (val), and 29,165.22 $\pm$ 3811.99 (test), a modest $\approx$ +2.8 % increase over baseline in testing. CNN yields 28,963.33 $\pm$ 4097.28, 29,037.79 $\pm$ 3997.60, and 28,991.57 $\pm$ 3798.61, aligning tightly with the baseline at $\approx$ +2.2 %. CNN-CT is also stable at 28,622.99 $\pm$ 4147.04, 28,718.03 $\pm$ 3913.19, and 29,050.75 $\pm$ 4127.60, though it overshoots slightly at testing. By contrast, LSTM (28,795.38 $\pm$ 4289.85, 28,959.75 $\pm$ 4220.90, 29,353.30 $\pm$ 4085.12) and GRU-CT (28,704.46 $\pm$ 4011.13, 28,858.93 $\pm$ 3959.79, 29,367.40 $\pm$ 4918.38) overshoot more strongly by $\approx$ +3.4 %. The weakest generalization is observed for CONV-CT at 28,646.79 $\pm$ 4100.34, 28,808.03 $\pm$ 4092.43, and 29,714.10 $\pm$ 3950.64, overshooting baseline by $\approx$ +4.7 %.

For price forecasting, the baseline (TRUE) registers 54.36 $\pm$ 15.00 in training, 59.30 $\pm$ 12.27 in validation, and 74.70 $\pm$ 11.46 in testing. CNN-CT delivers the strongest stability with 56.25 $\pm$ 13.10, 60.31 $\pm$ 11.04, and 69.15 $\pm$ 10.02 in testing, $\approx$ -7.4 % below baseline. XGB is comparably strong with 54.36 $\pm$ 13.55, 58.60 $\pm$ 9.97, and 69.39 $\pm$ 10.31 in testing ($\approx$ -7.1 %). LSTM remains competitive at 55.87 $\pm$ 11.99, 59.26 $\pm$ 9.13, and 70.23 $\pm$ 10.42, a $\approx$ -6.0 % improvement. GRU-CT, at 55.05 $\pm$ 13.06, 58.58 $\pm$ 9.98, and 71.24 $\pm$ 10.97, performs weaker ($\approx$ -4.6 %), while CONV-CT (55.98 $\pm$ 12.29, 59.83 $\pm$ 9.94, 72.21 $\pm$ 11.06) shows limited benefit ($\approx$ -3.3 %). CNN overshoots most strongly, with 54.40 $\pm$ 12.62, 57.29 $\pm$ 9.68, and 72.70 $\pm$ 13.59, a marginal $\approx$ -2.7 % gain.

Two limitations temper generalization. First, our evaluation used point error metrics; operational deployment requires uncertainty awareness. A variational heteroscedastic Gaussian process approach (Kou et al., 2015), produced calibrated prediction intervals with good density scores (e.g., NLPD $\approx$ 0.66–0.75 and |RE| $\approx$ < 4 % in the Australian market) while keeping MAPE within 3.7–6.1 % (Kou et al., 2015), suggesting that interval quality should complement RMSE/MAE reporting. Second, results are Spain-specific; cross-market performance can shift with market design and generation mix, as seen in (Lisi and Shah, 2024) and (Su et al., 2022).

Implications for the energy sector are clear. LSTM model offers dependable baselines for routine operations where stable median errors are preferred. GRU-CT and CNN-CT models are attractive for peak-sensitive applications (hedging, reserve scheduling) but require safeguards against overshoot. The CONV-CT model can be favored in short-horizon price desks but is less suitable for load operations that demand memory. Future work should quantify uncertainty alongside point accuracy using coverage and sharpness criteria, expose structural comovements via graph features to stabilize gated models, and explore frequency-aware hybrids that allocate modes to specialized learners. Extending evaluation beyond Spain and across seasons will clarify transferability and operational robustness.

5. Conclusion, limitations, and future directions

This study developed and validated a hybrid deep learning framework for day-ahead electricity load and price forecasting in the Spanish market. The proposed framework integrated temporal and seasonal

dynamics using LSTM, CNN, GRU, and temporal-context hybrid models, and incorporated exogenous drivers such as weather and calendar effects.

The results demonstrated that the LSTM model was most reliable for load forecasting, achieving the lowest MAE (2895.31 MWh) and MedAE (1796.77 MWh), while maintaining competitive RMSE performance against XGB. In contrast, for price forecasting, hybrid models performed better, with the CONV-CT model achieving the lowest RMSE (6.36 €/MWh) and MedAE (4.94 €/MWh). These findings confirm that while baseline models excel in minimizing absolute errors for load, hybrid temporal-context architectures offer clear advantages in capturing complex price dynamics and reducing forecasting errors.

The main contribution of this study lies in the development of a feature-rich, hybrid deep learning framework that jointly models load and price, explicitly addressing their interdependence and improving predictive accuracy under volatile and seasonal conditions. This contribution goes beyond previous studies by combining multiple architectures with contextual encoding while validating across multiple model types.

Nevertheless, certain limitations remain: the reliance on four years of Spanish data limits direct generalizability; hybrid models are computationally demanding and sensitive to hyperparameter tuning; and some real-time market drivers (e.g., fuel prices, cross-border flows) were not included.

Future research should extend this framework across multiple electricity markets and longer time horizons, integrate additional exogenous and real-time variables, and explore probabilistic and ensemble-based methods to better quantify uncertainty. Further efforts are also needed to reduce computational complexity and enhance interpretability, enabling broader adoption in real-time operational environments.

CRedit authorship contribution statement

Ahmed Adil Nafea: Methodology, Validation, Formal analysis, Investigation, Writing – review & editing. **Omer A. Alawi:** Conceptualization, Data curation, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Ziaul Haq Doost:** Methodology, Validation, Formal analysis, Investigation, Writing – review & editing, Visualization. **M.M. AlAni:** Methodology, Validation, Formal analysis, Investigation, Writing – review & editing. **Ahmad Bilal Ahmadullah:** Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Ravinesh C. Deo:** Resources, Supervision, Methodology, Writing – original draft, Writing – review & editing. **Zaher Mundher Yaseen:** Project administration, Resources, Supervision, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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